

JKBOSE 2026

CLASS 11TH

#XUV 

Happy Birthday

REDOX REACTIONS

LECTURE-04

CHEMISTRY

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JK Bole Wallah

Topics To Covered

- 1** **Balancing The Redox Reactions**
- 2** **Application The Redox Reactions**

Oxidation No. Method.

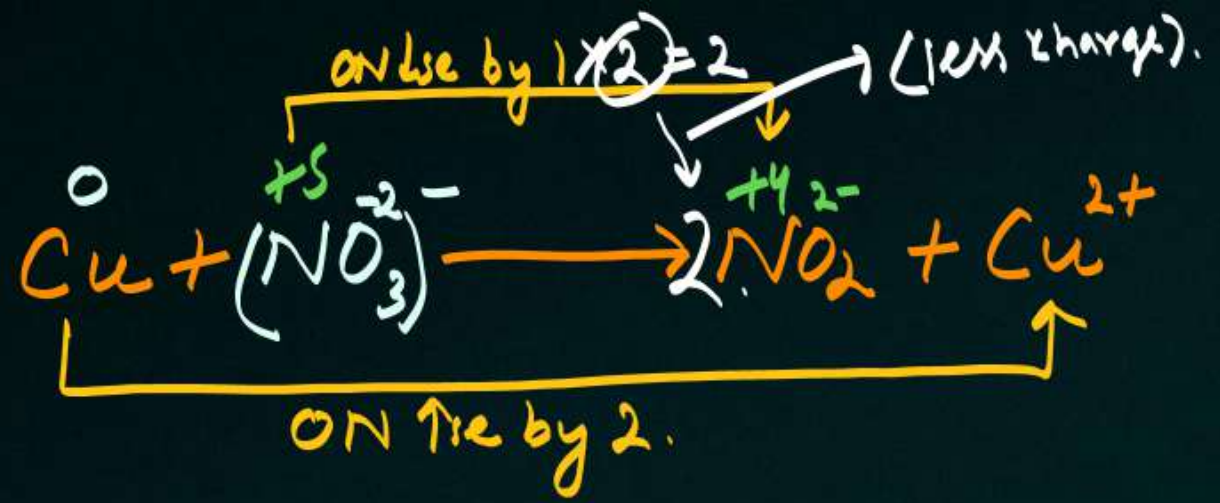
- 1) Write the skeletal equation (Unbalanced equation)
 - 2) Write the ON of each element.
 - 3) Calculate the change in ON & then balance it.
 - 4) Balance all atoms except H & O.
- 3) i) If acidic medium $\rightarrow$
- Balance 'O' atom by adding equal no. H_2O .
 - Balance 'H' atom by adding H^+ .

ii) Basic medium $\rightarrow$

Balance 'O' atom by adding equal no. of H_2O .

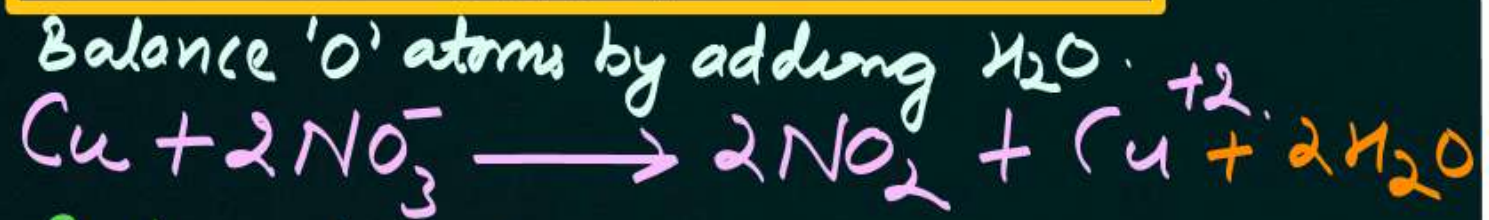
• Balance 'H' atoms.

first add ! no. of water molecules equal to the deficiency of hydrogen and equal no. of OH^- ions are added to the opposite side.

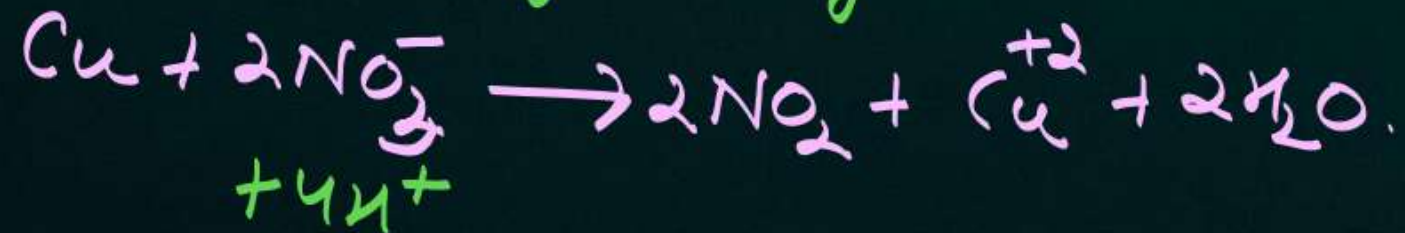


- Balance all atoms except H & O.
- $$\text{Cu} + 2\text{NO}_3^- \longrightarrow 2\text{NO}_2 + \text{Cu}^{+2}$$

ACIDIC MEDIUM.



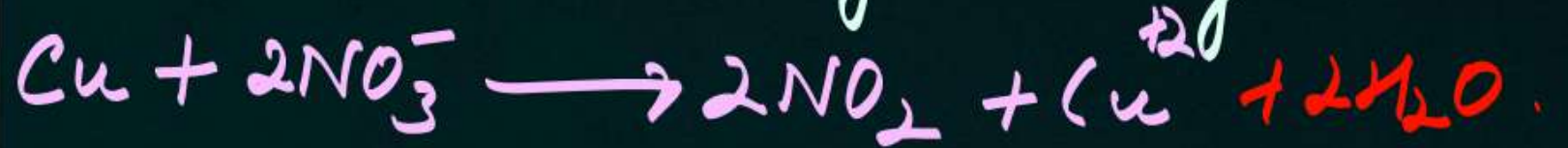
Balance 'H' by adding H^+



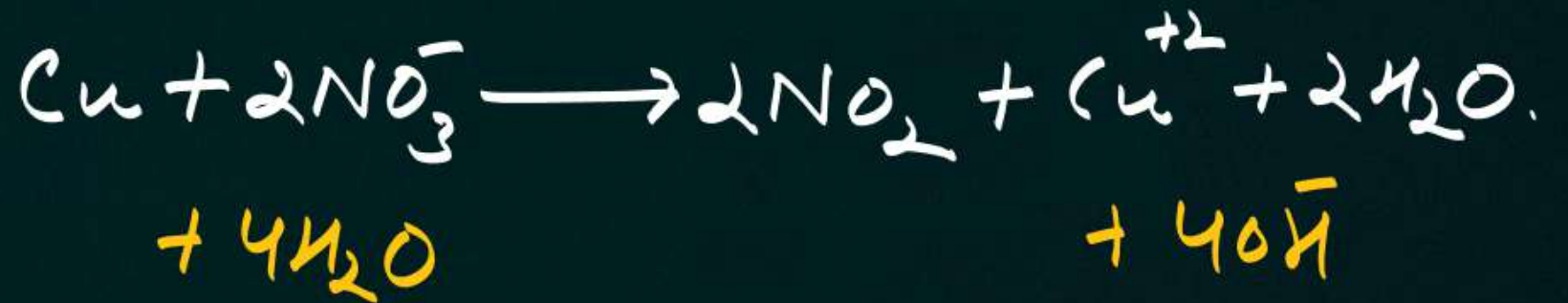
Hence rxn is balanced.

Basic Medium.

- Balance 'O' atoms by adding H_2O .



- Balance 'H' atoms by adding equal no. H_2O molecules & to the other side equal no. of OH^- is added.



Reaction is balanced.

Balanced Chemical Reactions

- ❖ The reaction in which no. of particular elements or atom. Present in reactant are equal to no. of element present in a product is called balanced **chemical reaction**.
- ❖ **Two Method**
 - (a) Oxidation Number Method
 - (b) Ion- e^- method or Half Rxn Method

Balancing The Redox Reactions

(A) Oxidation Number Method

The various steps involved in the balancing of redox equations by oxidation number method are:

Step 1. Write the skeletal equation of all the reactants and products of the reaction.

Step 2. Indicate the oxidation number of each element above its symbol and identify the elements which undergo a change in the oxidation number (O.N.).

Step 3. Calculate the increase or decrease in oxidation number per atom and equalise it by multiply with suitable number.

Step 4. Balance all atoms other than H and O.

Step 5. Finally balance H and O atoms by adding H_2O molecules using hit and trial method.

Balancing The Redox Reactions

Step 6. In case of ionic reactions,

- (a) **For acidic medium.** First balance O atoms by adding H_2O molecule to whatever side deficient in O atoms and then balance H atoms by adding H^+ ions to whatever side deficient in H atoms.
- (b) **For basic medium.** First balance O atoms by adding H_2O molecules to whatever side deficient in O atoms. The H atoms are then balanced by adding H_2O molecules equal in number to the deficiency of H atoms and an equal number of OH^- ions are added to the opposite side of the equation.

QUESTION

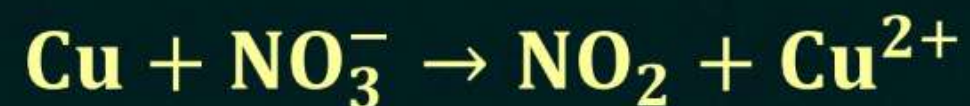


Write the balanced ionic equation for the reaction of potassium dichromate (VI), $\text{K}_2\text{Cr}_2\text{O}_7$, with sodium sulphite, Na_2SO_3 in acid medium to give chromium(II) ion and sulphate ion.

QUESTION

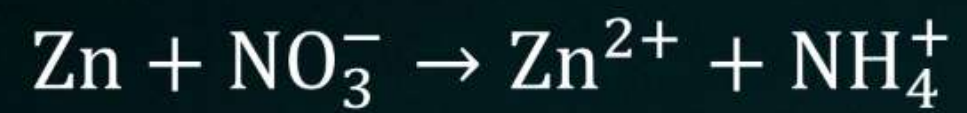


Copper reacts with nitric acid. A brown gas is formed and the solution turns blue.
The equation may be written as:



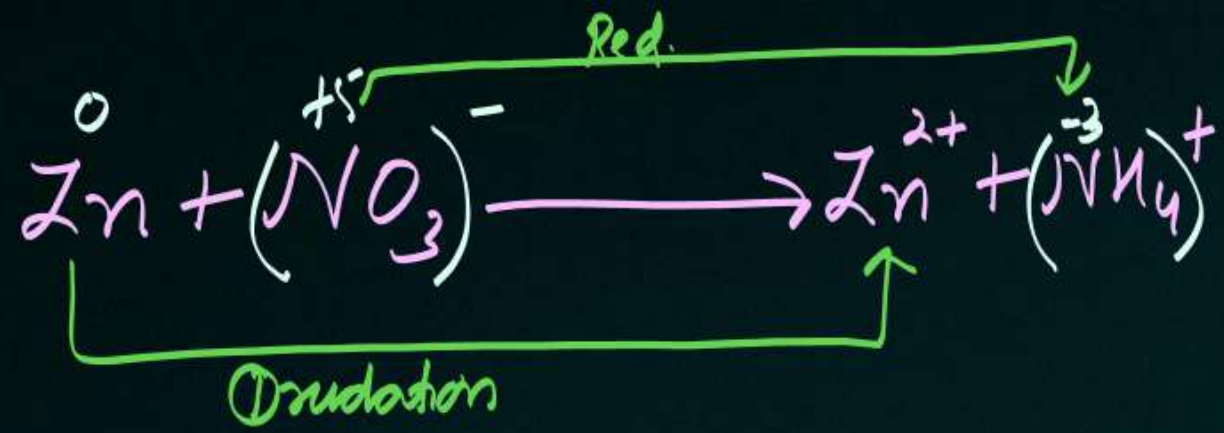
Balance the equation by oxidation number method.

QUESTION

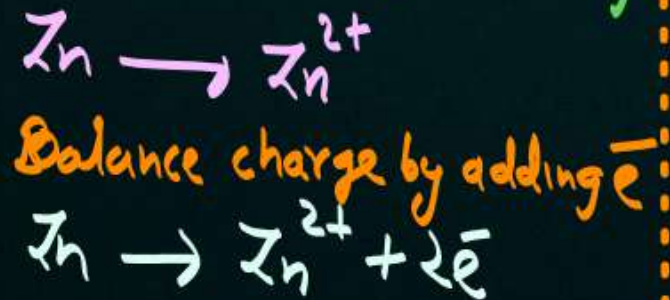
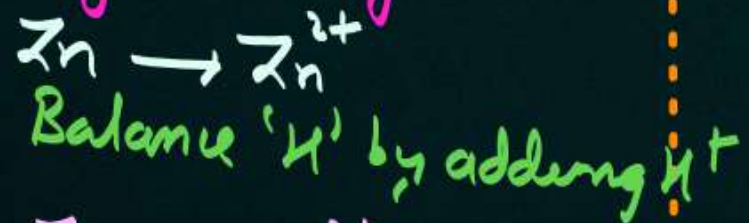
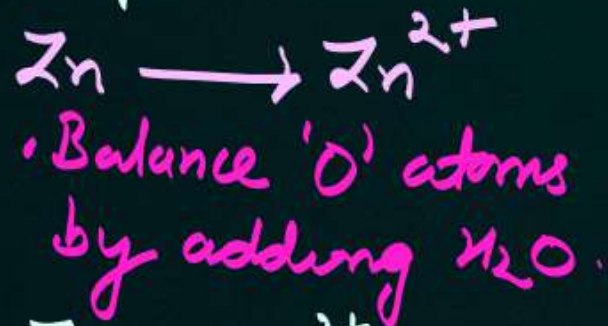
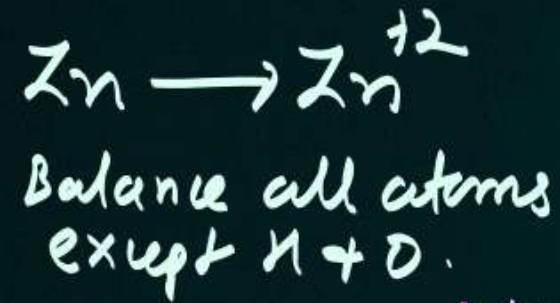


(in basic medium)

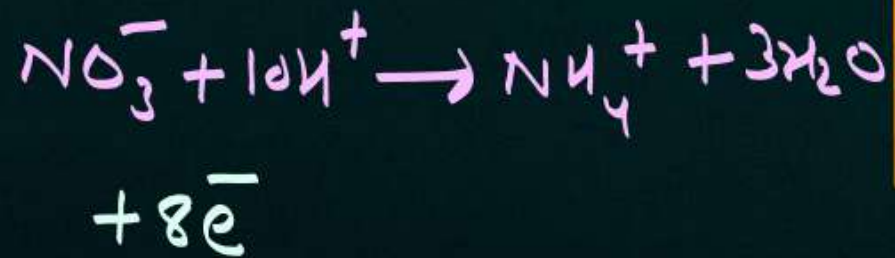




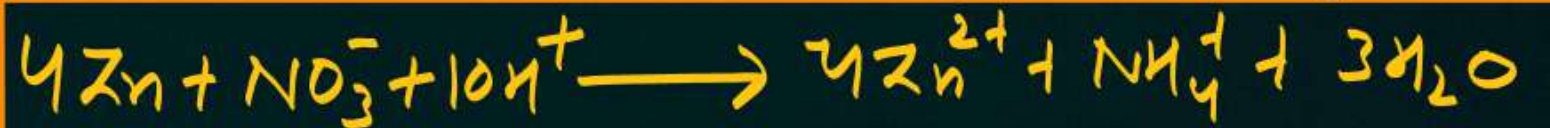
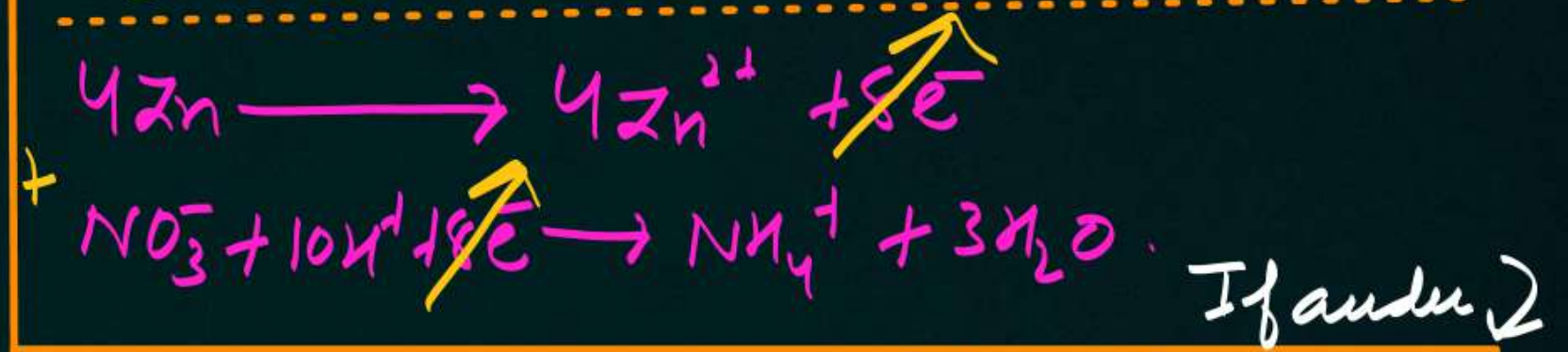
BREAK The Equation
Oxidation



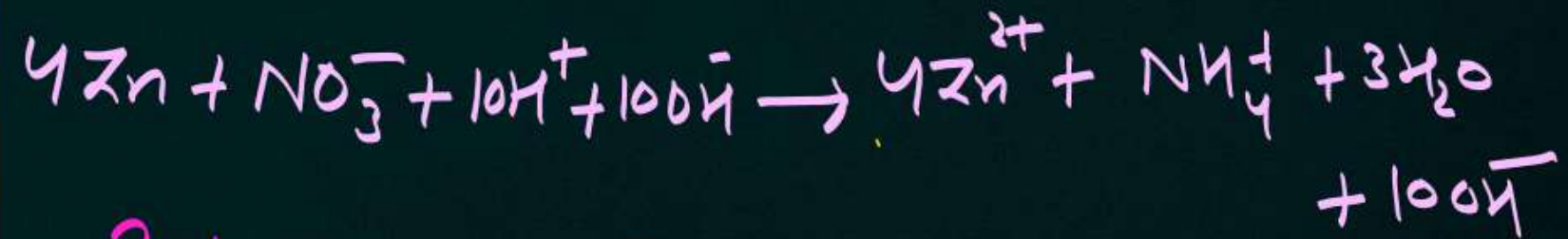
Reduction half



Adding the rxn so that e^- cancel.

$$(\text{Zn} \longrightarrow \text{Zn}^{2+} + 2e^-) \times 4$$


If Basic Medium



Balanced equation.

Balancing The Redox Reactions

(B) Ion-Electron Method (Balancing Equations Using Half Reactions)

1. Write the ionic form of Chemical Equation.
2. Identify the elements undergoing Oxidation & Reduction.
3. Break the Reaction into two Halves.
 - (i) Oxidation Half
 - (ii) Reduction Half
4. Balance the half reaction
 - (i) Balance all the atoms except 'H' & 'O'.
 - (ii) Balance the Oxygen by adding H_2O Molecule to the Side having less no. of oxygen.
 - (iii) Balance to hydrogen by adding H^+ to the Side having less no. Of Hydrogen.
 - (iv) Balance the charge by adding electron.
5. Add the two half Such that electron cancel each other.
6. In Basic Medium add as many OH^- ions to the both Side of reaction as H^+ were added previously.

If Acidic
Med.

If basic.

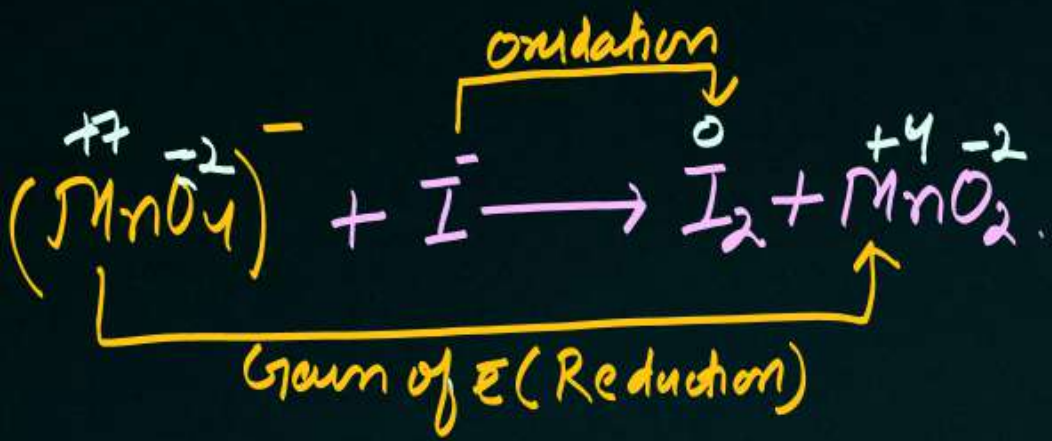
QUESTION



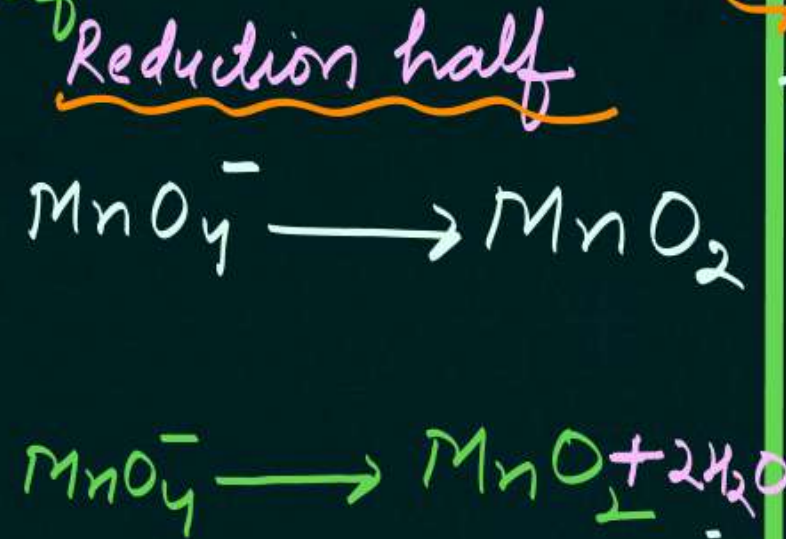
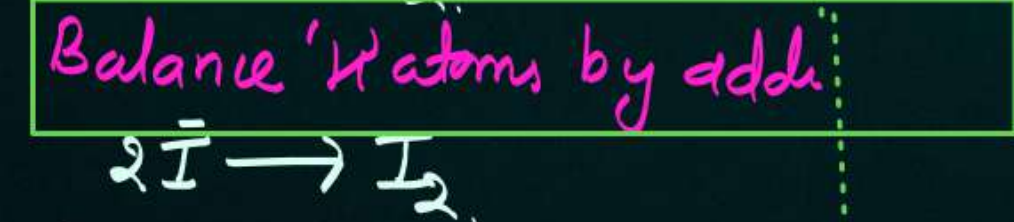
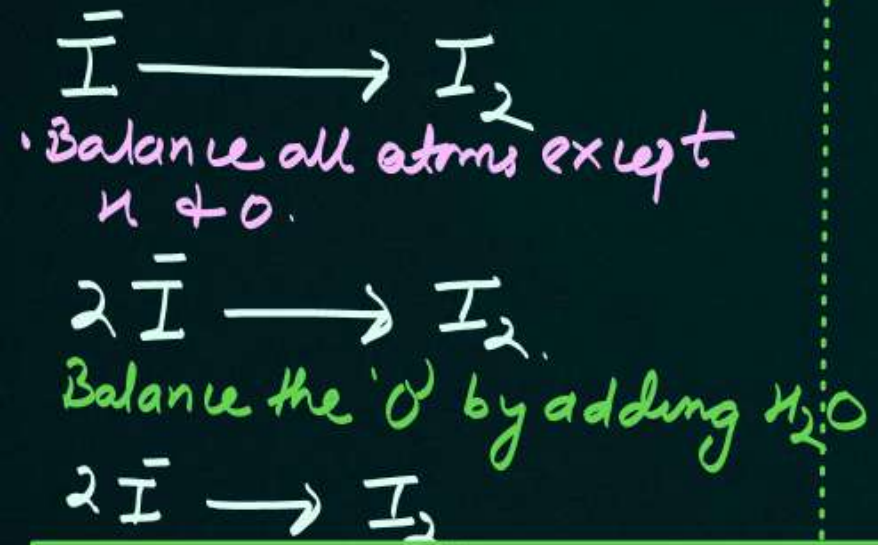
Permanganate (VII) ion, MnO_4^- in basic solution oxidises iodide ion, I^- to produce molecular iodine (I_2) and manganese (IV) oxide (MnO_2). Write a balanced ionic equation to represent this redox reaction.

Sol: $\rightarrow$

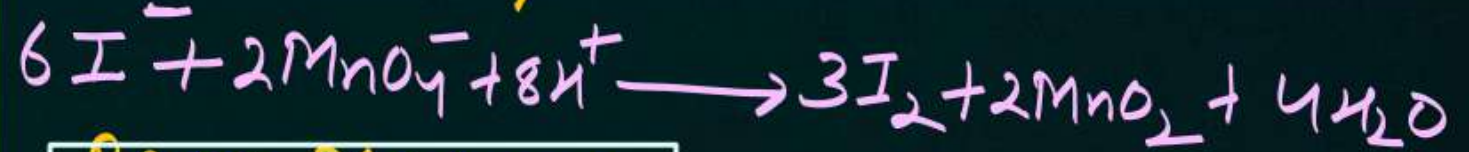
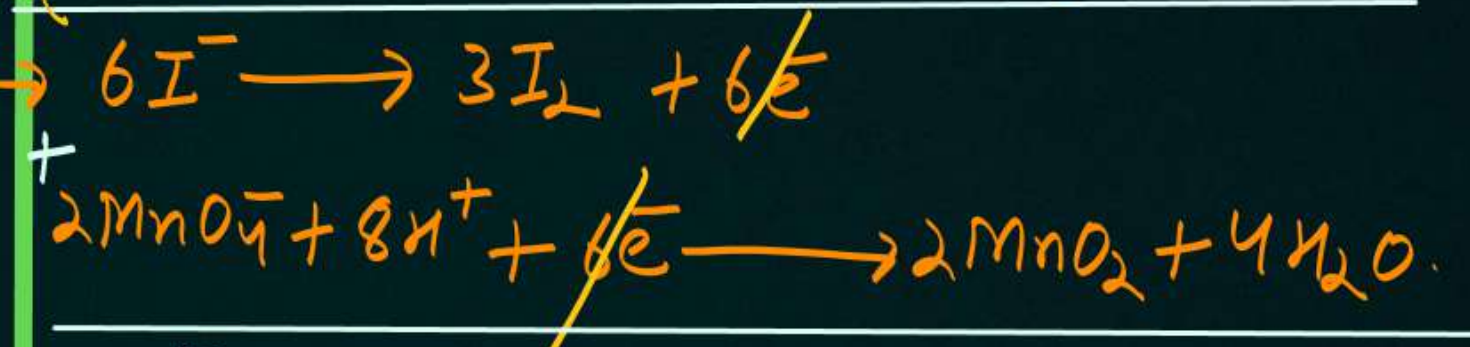
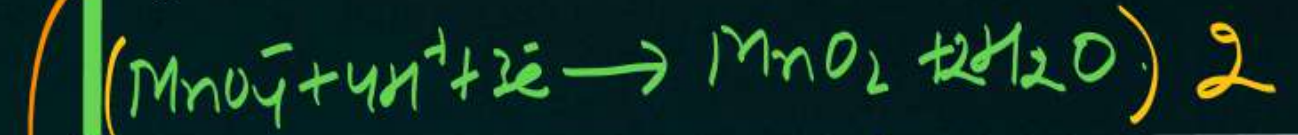
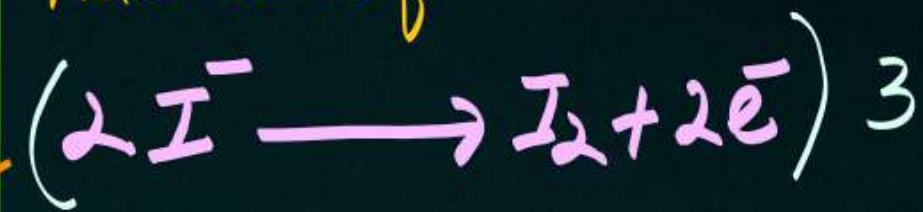




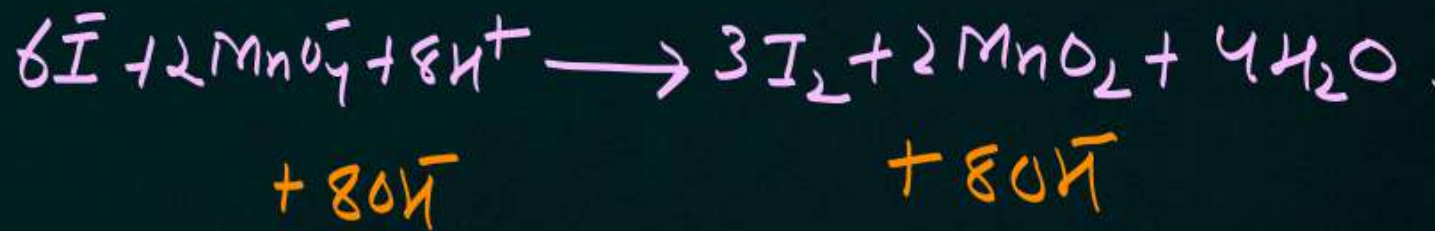
BREAK The equation in two half



Addition of Oxidation & Reduction half



BASIC MEDIUM.

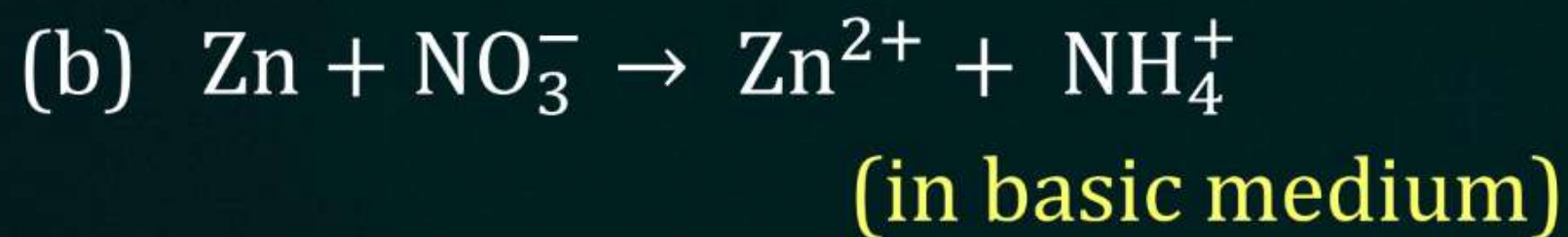
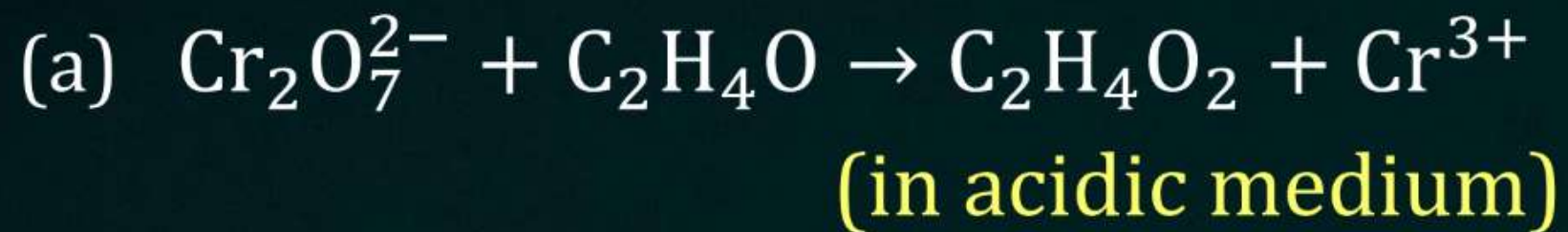


Reaction is Balanced.

QUESTION



Balance the following equations by ion electron method:



QUESTION

ON method.
OR

Balance the following redox reactions by ion-electron method:

- (a) $\text{MnO}_4^- (\text{aq}) + \text{I}^- (\text{aq}) \rightarrow \text{MnO}_2 (\text{s}) + \text{I}_2 (\text{s})$ (in basic medium)
- (b) $\text{MnO}_4^- (\text{aq}) + \text{SO}_2 (\text{g}) \rightarrow \text{Mn}^{2+} (\text{aq}) + \text{HSO}_4^- (\text{aq})$ (in acidic solution)
- (c) $\text{H}_2\text{O}_2 (\text{aq}) + \text{Fe}^{2+} (\text{aq}) \rightarrow \text{Fe}^{3+} (\text{aq}) + \text{H}_2\text{O} (\text{l})$ (in acidic solution)
- (d) $\text{Cr}_2\text{O}_7^{2-} + \text{SO}_2 (\text{g}) \rightarrow \text{Cr}^{3+} (\text{aq}) + \text{SO}_4^{2-} (\text{aq})$ (in acidic solution)

QUESTION

Choose the correct explanation regarding half-reaction such as $\text{Cr}_2\text{O}_7^{2-} \longrightarrow \text{Cr}^{3+}$ from the following.

- 1 It is oxidation half-reaction
- 2 Chromium being oxidised
- 3 $\text{Cr}_2\text{O}_7^{2-}$ has a good reducing agent
- 4 Chromium being reduced

QUESTION

An oxidation process involves:

- 1 Increase in oxidation number
- 2 Decrease in oxidation number
- 3 Addition of electrons
- 4 Removal of electronegative element

QUESTION



- (a) Define oxidation and reduction on the basis of electron transfer concept. Give example in each case.
- (b) In the reactions given below, identify the species undergoing oxidation and reduction:

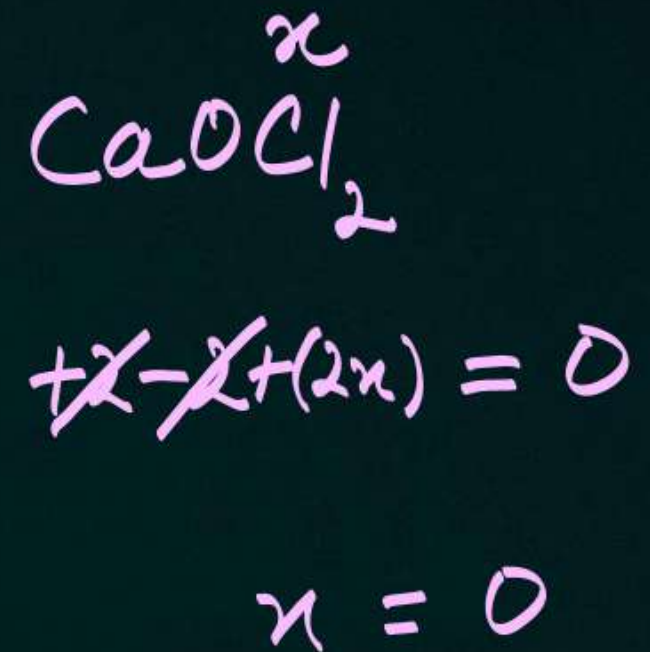
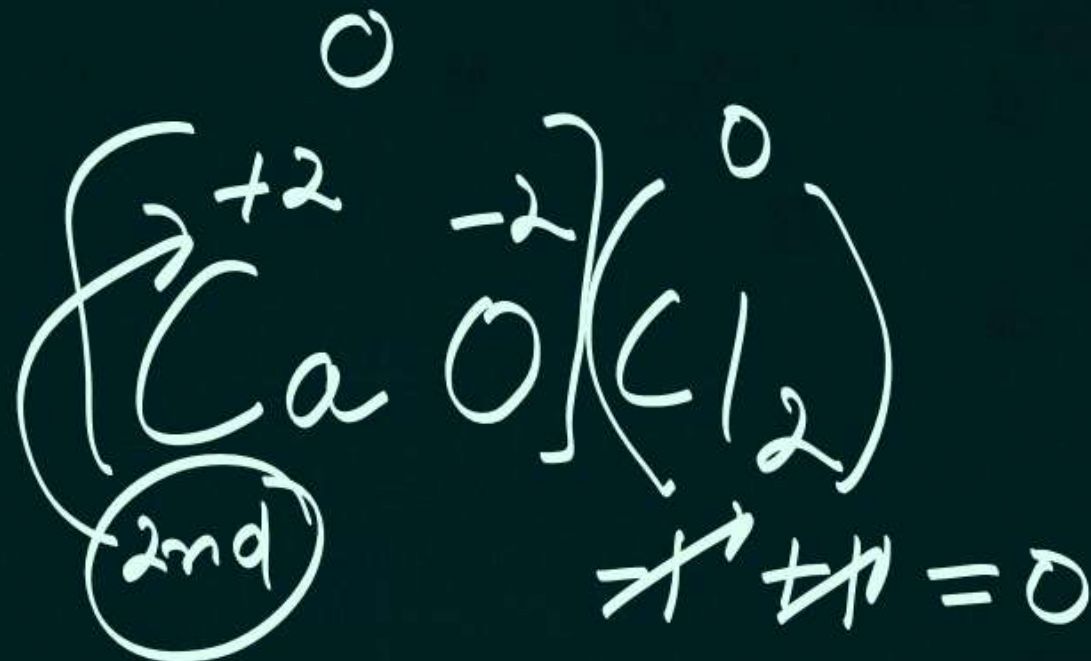


(2024)

3 Marks

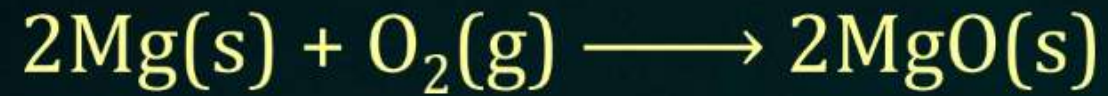
b) $+1, -1$

d) 0, -1



QUESTION

In the following reaction,



with respect to Mg, the process is called

- 1 Oxidation
- 2 Reduction
- 3 Redox reaction
- 4 None of the above

QUESTION

Identify disproportionation reaction.



Homework



NEVER MISS THE Lec.
of Organic Chemistry.

6 - July - 2020

Thank
You

SO MUCH FOR MAKING
my Birthday Special.